



AQ1.4: Activity Questions 4 - Not Graded



This assignment will not be graded and is only for practice.

4.1 Level 1:

Let A be a 3×3 matrix with non-zero determinant. If $\det(2A) = k \det(A)$, then what will be the value of k ?

[Hint: If a scalar (c) is multiplied with one row of a matrix A , then the determinant of the new matrix will be c times the determinant of A .]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 8

1 point

1 point

If $A = \begin{bmatrix} 2 & -11 & 1 \\ 1 & -3 & 4 \\ -1 & 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 13 & -2 & 0 \\ 0 & 1 & -1 \\ 3 & 4 & 2 \end{bmatrix}$, then choose the set of

correct options.

☐

$\det(A) = -9$ and $\det(B) = -3$

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$\det(A) = 9$ and $\det(B) = 3$

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$\det(A) = -9$ and $\det(B) = 3$

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$\det(AB) = -27$ and $\det(BA) = 27$

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$\det(AB) = \det(BA) = -27$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\det(A) = -9$ and $\det(B) = 3$

$\det(AB) = \det(BA) = -27$

1 point

Let A be a 2×2 matrix, which is given as $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$. Define the following

matrices :

$B = \begin{bmatrix} a_{11} - a_{21} & a_{12} - a_{22} \\ a_{21} & a_{22} \end{bmatrix}$, $C = \begin{bmatrix} a_{11} - a_{21} & a_{12} - a_{22} \\ a_{21} & a_{22} \end{bmatrix}$

$D = \begin{bmatrix} a_{11} + a_{21} & a_{12} - a_{22} \\ a_{21} & a_{22} \end{bmatrix}$, $E = \begin{bmatrix} a_{11} - a_{21} & a_{12} + a_{22} \\ a_{21} & a_{22} \end{bmatrix}$



Which of the matrices among B, C, D , and E have the same determinant as that of the matrix A , for any real numbers $a_{11}, a_{12}, a_{21}, a_{22}$?

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BB and DD

☐

BB and EE

☐

BB and CC

☐

DD and EE

☐

CC and EE

No, the answer is incorrect.

Score: 0

Accepted Answers:

BB and CC

Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ r a_{31} & r a_{32} & r a_{33} \end{bmatrix}$ be a matrix and $r, s, t \neq 0$. Find $\det(A)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

Suppose $A = \begin{bmatrix} -14 & -2 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 & -1 \\ 4 & \alpha & 0 \\ 0 & -3 & -5 \end{bmatrix}$. If

$\det(AB) = 32$, then find the value of α .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

4.2 Level 2:

(Use this information to answer questions 6 and 7)

Let $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ be the vertices of a

triangle. The area of the triangle ABC is given by $\frac{1}{2}|\det(D)|$ square units, where

$$D = \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}$$

If $P(-1, 1)$, $Q(1, 1)$ and $R(1, 0)$ are the vertices of a triangle, then find the area (in square units) of the triangle PQR .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

1 point

Let $P(x-1, 2x)$, $Q(0, x-2)$ and $R(-x+1, 1)$ be the vertices of a triangle. If the area of the triangle is 55 square units and $x > 0$, then choose the set of correct options

☐

length of side $PQ = \sqrt{29}$

☐

units.

☐

length of side $PR = \sqrt{40}$

☐

units.

☐

length of side $QR = 2$

☐

length of side $QR = 1$

No, the answer is incorrect.

Score: 0

Accepted Answers:

length of side $PQ = \sqrt{29}$

☐

units.

length of side $QR = 2$

1 point

Find out the correct value of $\det(A)$, where $A = \begin{bmatrix} 120102020 \times 2030 \\ 120202030 \times 2010 \\ 120302010 \times 2020 \end{bmatrix}$

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[Hint: Observe that the given matrix is of the form: $\begin{bmatrix} 1abc \\ 1bca \\ 1cab \end{bmatrix}$]

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☐

200200

☐

-2000-2000

☐

20002000

No, the answer is incorrect.**Score: 0****Accepted Answers:**

20002000

1 point

Let AA be a square matrix of order 33 and BB be a matrix that is obtained by adding 22 times the first row of AA to the third row of AA and adding 33 times the second row of AA to the first row of AA . What is the value of $\det(6A^2 B^{-1})\det(6A2B-1)$?

☐ $6 \det(A)6 \det(A)$ ☐ $6 \det(A)\det(B)6 \det(A)\det(B)$ ☐ $6^3 \det(A)63 \det(A)$ ☐ $6^3 \det(A)\det(B)63 \det(A)\det(B)$ **No, the answer is incorrect.****Score: 0****Accepted Answers:** $6^3 \det(A)63 \det(A)$

Suppose $A = \begin{bmatrix} 192021 \\ 202119 \\ 211920 \end{bmatrix}$. Find the value of $\frac{1}{60} \det(A^2)601 \det(A2)$.

[[You can try to compute the determinant of a matrix of the form: $\begin{bmatrix} abc \\ bca \\ cab \end{bmatrix}$]]

No, the answer is incorrect.**Score: 0****Accepted Answers:**

(Type: Numeric) 540

1 point